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Lab 1: Implementation of Source Code Management Using Git Commands Through Github

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Lab Report

Objective

To understand and apply Source Code Management (SCM) using Git commands and GitHub for version control and project management.

Tools Required

- Git
- GitHub Account
- Git Bash / Terminal
- Visual Studio Code (Optional)

Theory

Source Code Management (SCM) is the process of tracking and managing changes to source code. Git is a distributed version control system that records changes, while GitHub is a cloud-based platform for hosting and collaborating on Git repositories.

Procedure

1. Install Git and configure the username and email.
2. Create a project folder and initialize a Git repository.
3. Add project files to the staging area.
4. Commit the changes to the local repository.
5. Create a new repository on GitHub.
6. Link the local repository to the GitHub repository.
7. Push the local repository to GitHub.

Git Commands (Coding Section)

```
# Configure Git
   git config --global user.name "Your Name"
   git config --global user.email
"your@email.com"

# Create and initialize repository
```

```
  mkdir MyProject
  cd MyProject
  git init

# Create a sample file
echo "Hello GitHub!" > README.md

# Stage and commit changes
git add .
git commit -m "Initial commit"

# Connect to GitHub repository
git remote add origin
https://github.com/username/MyProject.git

# Push project to GitHub
git branch -M main
git push -u origin main

# Check repository status
git status

# View commit history
git log
```

Output

- A local Git repository was successfully created.
- Files were tracked and committed.
- The repository was connected to GitHub.
- The project was successfully pushed to the remote GitHub repository.

Result

Successfully implemented Source Code Management using Git commands and synchronized the local repository with GitHub.

Conclusion

This lab demonstrated the complete Git workflow, including repository initialization, staging, committing, connecting to GitHub, and pushing code. It provided practical experience with version control and collaborative software development.